Neil Patel

US Citizen | ndpatel@unc.edu | linkedin.com/in/neil-patel-735132281 | github.com/NeilP211

Education

University of North Carolina at Chapel Hill — Expected May 2028
Bachelor of Science in Computer Science — *Chapel Hill, NC*
- Courses: Data Structures & Algorithms, Computer Systems (C), Object-Oriented Design (Java), Linear Algebra

Experience

ASEA Brown Boveri (ABB) — Cary, NC
Machine Learning Engineer Intern — *Aug. 2026 - Present*
- Building the evidence case to switch on a dormant embedding-similarity routing tier in production: expanding the golden set with paraphrased, multilingual & adversarial queries to measure what it catches that rules miss
- Instrumenting per-agent latency, retry, failure-mode & token-cost metrics across an 8-agent LLM pipeline

Software Engineering Intern — *May 2026 - Aug. 2026*
- Engineered a real-time parts-replenishment platform on AWS replacing manual inventory-tracking across 5 production lines, recovering an estimated 900K parts/year ($9M in production value) by eliminating material-starvation
- Integrated it into an existing Kepware, Tulip and AWS IoT Core MQTT telemetry pipeline, serving inventory and lead-time state out of RDS and SQL through a Python backend to a React dashboard
- Owned CI/CD pipelines deploying 2 interdependent services on every merge, removing manual release steps

SoberRide (Startup) — Oct. 2024 - May 2025
Director of Engineering — *Raleigh, NC*
- Co-architected a non-invasive facial intoxication classifier by fine-tuning an ImageNet-pretrained VGG-16 (Adamax) on 10K Haar-cascade-cropped face images, reaching 83% accuracy and beating a GPT-4o vision baseline
- Secured a U.S. patent; drove 400K+ views at CES 2025 (youngest startup), featured in IEEE Spectrum, won the Conrad/Diamond Challenge, endorsed by Mothers Against Drunk Driving, backed by Rep. Ross & Gov. Stein

NC State University — Mar. 2023 - Mar. 2025
Lead Research Intern — *Raleigh, NC*
- Published IEEE paper on skin cancer classification for minority demographics, increasing accuracy from 82% to 89%
- Roughly tripled the skin-tone-skewed dataset to Fitzpatrick IV-VI by masking lesions with a U-Net/R2U-Net+CBAM segmenter (0.79 IoU), engineering style transfer to background skin while preserving diagnostic detail

Projects

distkv | *Go, gRPC, Raft, bbolt, Kubernetes, Docker* — github.com/NeilP211/distkv
- Built a distributed key-value store on a from-scratch Raft engine: leader election, log replication, snapshots
- Served 9.9K linearizable reads/s at 5 ms P95 via ReadIndex on a 3-node cluster, with 820 ms leader failover
- Built a deterministic network simulator for partition, drop, delay and crash/restart that replays all 5 seeded chaos scenarios exactly in CI under the race detector, alongside unit tests on every Raft subsystem
- Verified correctness over 152K ops in 60s of seeded fault-injection tests: 100% linearizable, 5 chaos scenarios
- Shipped 5 Prometheus metrics + a Grafana monitoring dashboard, a 3-replica Kubernetes StatefulSet + Helm chart

Crypt | *Rust, axum, gRPC, PostgreSQL, Redis, Python, Docker* — github.com/NeilP211/crypt
- Served 10,959 places through microservices: a Rust axum REST and gRPC API over PostgreSQL/PostGIS and Redis plus a Python FastAPI CLIP embedding service, booted by Docker Compose
- Built a from-scratch Rust HNSW vector index hitting 97.2% recall@10 vs exact kNN at P99 4.9 ms over 100k vectors
- Ranked 6x over-fetched candidates by blended vector, geo and metadata scores (0.6/0.3/0.1), cached in Redis

Exfil | *Go, Rust, TypeScript, React, WebSocket* — github.com/NeilP211/exfil
- Built a real-time, server-authoritative browser shooter on a 64Hz Go simulation, sustained with zero tick drops
- Cut snapshots to a 93-byte P95 at 6 players with delta compression over a binary WebSocket network protocol
- Mirrored the sim in Rust (Q16.16 fixed-point); CI proves both produce identical world hashes over 2000 ticks

Publications

S. Chilaka, N. Patel, "Pass@k Is Not a Property of a Model," *Preprints*, 2026.
Kalra, Iyer, N. Patel, "Enhancing Skin Cancer Detection Across Skin Tones with Style Transfer Augmentation," *ACIT 2024*.
N. Patel, in G. Friedland (Ed.), *Information-Driven Machine Learning*, *Springer Nature*, 2024.

Technical Skills

Languages: Python, Java, C, TypeScript, JavaScript, SQL Certifications: AWS Certified Cloud Practitioner
Infrastructure: AWS, Docker, Kubernetes, Helm, Terraform, Linux Tools: Git, CI/CD, unit tests, Claude Code
Backend: gRPC, REST, WebSocket, Raft consensus Storage: PostgreSQL, Redis, bbolt Frontend: React, Next.js
Monitoring: Prometheus, Grafana AI & ML: PyTorch, RAG, LLM agents, LLM-judge evals